Quantum Blackholes, Wall Crossing and Mock Modular Forms

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DECLARATION

I, **Tharun Naagesh (Roll No: IMS22252)**, hereby declare that, this report entitled **"Quantum Blackholes, Wall Crossing and Mock Modular Forms"** submitted to Indian Institute of Science Education and Research Thiruvananthapuram towards the partial requirement of **Master of Science** in **Mathematics**, is an original work carried out by me under the supervision of **Dr. Bindusar Sahoo** and has not formed the basis for the award of any degree or diploma, in this or any other institution or university. I have sincerely tried to uphold academic ethics and honesty. Whenever a piece of external information or statement or result is used then, that has been duly acknowledged and cited.

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CERTIFICATE

This is to certify that the work contained in this project report entitled **"My Project Title"** submitted by [**Full Name**] (**Roll No:** [**Roll Number**]) to Indian Institute of Science Education and Research, Thiruvananthapuram towards the partial requirement of [**Master of Science/ Doctor of Philosophy**] in [**Department Name**] has been carried out by [him/her/them] under my supervision and that it has not been submitted elsewhere for the award of any degree.

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April 8, 2026 Project Supervisor

ABSTRACT

If you have to structure it as objective, methods, results, and conclusions, do it that way.

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Chapter 1

Prelimnaries

1.1 Primer on Fourier Series

1.1.1 Fourier Series

Definition 1.1.1. ($\mathbb{S}$) Denote $\mathbb{S}$ as the space of all complex valued $\mathbb{Z}$ periodic functions on $\mathbb{R}$ that are Lebesgue Measurable (For the applications, we take them to be *continuous*). If $f \in \mathbb{S}$, then $\forall\ x \in \mathbb{R},\ f(x) = f(x+1)$

Definition 1.1.2. (Hermitian Inner Product) On $\mathbb{S}$ (Which is a vector space), define $\langle f, g\rangle := \int_0^1 f\bar{g}dx$

Definition 1.1.3. (Fourier Coefficients c_n and the Fourier Series) For $n \in \mathbb{Z}$

$$c_n(f) := \langle f, \exp(2\pi inx)\rangle = \int_0^1 f \exp(-2\pi inx)dx$$

The Fourier Series of the function f is defined as

$$\sum_{-\infty}^{\infty} c_n \exp(2\pi inx) = \sum_{n=-\infty}^{\infty} (\int_0^1 f(t)\exp(-2\pi int)dt)(\exp(2\pi inx))$$

Lemma 1.1.4. *Let $f : \mathbb{R} \to \mathbb{C}$ be a $\mathbb{Z}$ periodic function. Let $c_n := \langle f, \exp(2\pi inx)\rangle$ as usual. Call f "nice" if $\{c_n\}$ converges abosolutely, $|c_n| \leq C/n^2$ for some constant C and for some $e > 1/2$, for all $x \in \mathbb{R}, h \in \mathbb{R}^+$,*

$$|f(x+h) - f(x)| \leq C|h|^e$$

(e=1 is called Lipshitz*).*
If f is "nice" and $c_n(f) = \langle f(x), \exp(2\pi inx)\rangle = 0$ for all $n \in \mathbb{N}$, then $f \equiv 0$.

Lemma 1.1.5. *If $f : \mathbb{R} \to \mathbb{C}$ (a $\mathbb{Z}$ periodic function) is "nice" (as defined in 1.1.4), then*

$$\sum_{n\in\mathbb{Z}} c_n \exp(2\pi inx) = f(x)$$

Lemma 1.1.6. ***(Parseval)*** *Suppose $f : \mathbb{R} \to \mathbb{C}$ is a "nice" function (as defined in 1.1.4). Then*

$$\langle f, f\rangle = \sum_{n\in\mathbb{Z}} |c_n|^2$$

Proof. $\langle f, f\rangle = \int_0^1 (\sum_{n\in\mathbb{Z}} c_n \exp(2\pi i n x))(\sum_{m\in\mathbb{Z}} c_m \exp(-2\pi m x))dx$ which gives

$$\sum_n \sum_n c_n c_m \int_0^1 (\exp(2\pi i x(n-m))dx)$$

which yields

$$\sum_n \sum_m c_n c_m \delta_{mn} = \sum_n |c_n|^2$$

□

1.1.2 (General Discussion) Periodic Functions of a Real Variable

Let $f : \mathbb{R} \to \mathbb{C}$ be $1-$periodic, i.e. $f(x+1) = f(x)$. This lifts to a unique function $\tilde{f} : S^1 \to \mathbb{C}$.

Constructing periodic functions

Let $g : \mathbb{R} \to \mathbb{C}$, and suppose g decays sufficiently fast at ∞. Let

$$f(x) := \sum_{n\in\mathbb{Z}} g(x+n)$$

This function can be easily checked to be 1 periodic.

Unfolding Trick, and Poisson Summation Formula

We have the simple fact

$$\int_{-\infty}^{\infty} g(x)dx = \sum_{n\in\mathbb{Z}} \int_0^1 g(x+n)dx$$

If f is periodic and is generated by g, then $f(x) = \sum_{n\in\mathbb{Z}} g(x+n)$ and

$$\int_0^1 f = \int_{-\infty}^{\infty} g$$

Fourier Coefficients of f:

$$f_m = \int_0^1 f(x)e^{-2\pi i m x}dx = \int_0^1 (\sum_{n\in\mathbb{Z}} g(x+n))e^{-2\pi i m x}dx$$

$$= \sum_{n\in\mathbb{Z}} \int_0^1 g(x+n)e^{-2\pi i m(x+n-n)}dx = \sum_{n\in\mathbb{Z}} \int_0^1 g(x+n)e^{-2\pi i m(x+n)}e^{2\pi i m n}dx$$

$$= (\text{ since } e^{2\pi i m n} = 1) \sum_{n\in\mathbb{Z}} \int_0^1 g(x+n)e^{-2\pi i m(x+n)}dx = \int_{-\infty}^{\infty} g(u)e^{-2\pi i m u}du$$

We thus have:

$$f_m = \hat{g}(m)$$

where $\hat{g}$ denotes the Fourier transform of g.

$$\boxed{f(x) = \sum_{n \in Z} g(x+n) = \sum_{m \in \mathbb{Z}} \hat{g}(m) e^{2\pi i m x}}$$

In the above equation, plugging $x = 0$ gives us the famed ***Poisson Summation Formula:***

Theorem 1.1.7. ***(Poisson Summation)***

$$\sum_{n \in \mathbb{Z}} g(n) = \sum_{m \in \mathbb{Z}} \hat{g}(m)$$

Theorem 1.1.8. ***(Basis for $L^2(S^1)$)*** *$e^{2\pi i n x}$ are periodic with period 1 and they form a basis for $L^2(S^1)$. Moreover:*

1. $$\int_0^1 e^{2\pi i n x} e^{-2\pi i m x} = \delta_{nm}$$

2. $$\sum_{\mathbb{Z}} e^{2\pi i m (x-y)} = \sum_{\mathbb{Z}} e^{2\pi i m x} e^{-2\pi i n x} = \sum_{m \in \mathbb{Z}} \delta(x - y + m)$$

   (Where $\delta(x-y)$ denotes the Dirac delta centered at y.)

Proof. We prove (2): Note that if $x = y$, we get ∞ in the LHS, and $\sum_m \delta(m) = \delta(0) = \infty$. If $x \neq y$, We apply the unfolding trick: Let $f(x-y) = \sum_{n \in \mathbb{Z}} \delta(x - y + n)$ (a periodic function).

$$f(\theta) = \sum_m f_m e^{2\pi i m \theta}$$

where

$$f_m = \int_0^1 (\sum_{n \in \mathbb{Z}} \delta(\theta + n)) e^{-2\pi i m \theta} d\theta$$

$$= \sum_{n \in \mathbb{Z}} \int_0^1 \delta(\theta + n) e^{-2\pi i m (\theta + n)} d\theta \text{ (since } e^{-2\pi i m x} \text{ is periodic)}$$

$$= \int_{-\infty}^{\infty} \delta(\theta) e^{-2\pi i m \theta} d\theta = 1$$

Hence, we see that the Fourier Coefficients of our f (defined using sum of δ) are 1. Hence

$$\sum_{n \in \mathbb{Z}} \delta(x - y + n) = \sum_{m \in \mathbb{Z}} f_m e^{-2\pi i m (x-y)} = \sum_{m \in \mathbb{Z}} e^{-2\pi i m (x-y)}$$

Which was what we set out to prove.

□

Cauchy's Theorem

Theorem 1.1.9. ***(Cauchy- weak version)*** *Let $f : U \to \mathbb{C}$ be holomorphic on the simply connected domain U. Let P be some polygon on U. Then*

$$\int_P f(z) dz = 0$$

1.2 Modular Forms

1.2.1 Basic Definitions and Results

Definition 1.2.1. (Upper Half Plane $\mathbb{H}$)

$$\mathbb{H} := \{z \in \mathbb{C} : im(z) > 0\}$$

Definition 1.2.2. ($SL(2,\mathbb{R})$)

$$SL(2,\mathbb{R}) := \left\{M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a,b,c,d \in \mathbb{R} \textbf{ and } ad - bc = 1\right\}$$

Definition 1.2.3. ($GL(2,\mathbb{R})$)

$$GL(2,\mathbb{R}) := \left\{M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a,b,c,d \in \mathbb{R} \textbf{ and } ad - bc \neq 0\right\}$$

Definition 1.2.4. ($GL^+(2,\mathbb{R})$)

$$GL^+(2,\mathbb{R}) := \left\{M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a,b,c,d \in \mathbb{R} \textbf{ and } ad - bc > 0\right\}$$

We have rather trivially the following inclusions

$$SL(2,\mathbb{R}) \hookrightarrow GL^+(2,\mathbb{R}) \hookrightarrow GL(2,\mathbb{R})$$

Definition 1.2.5. (Modular Form (for $SL_2(\mathbb{Z})$)) Let $k \in \mathbb{Z}$. $f : \mathbb{H} \to \mathbb{C}$ is a modular form of weight k if:

1. For all $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2,\mathbb{Z})$ and $z \in \mathbb{H}$,

$$f(\frac{az+b}{cz+d}) = (cz+d)^k f(z)$$

2. $f(z)$ is bounded as $im(z) \to \infty$
3. f is Holomorphic on all of $\mathbb{H}$

Corollary 1.2.6. $\forall x \in \mathbb{H}\ f(x+1) = f(x)$

Proof. Since $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \in SL(2,\mathbb{Z})$, we have the result trivially.

□

Corollary 1.2.7. $\forall x \in \mathbb{H}\ f(-1/x) = x^k f(x)$

Proof. Since $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in SL(2, \mathbb{Z})$, we have the result.

□

Corollary 1.2.8. $f(x) = (-1)^k f(x)$

Proof. Since $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \in SL(2, \mathbb{Z})$, the result holds. Moreover, if k is odd, f has to be the 0 map. ***The only non zero modular forms are of even weights.***

□

Fact 1.2.9. $SL(2, \mathbb{Z})$ is generated by $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

Corollary 1.2.10. *From 1.2.9 we have that any modular form can be re-characterised (without reference to $SL(2, \mathbb{Z})$) by the following axioms: $f : \mathbb{H} \to \mathbb{C}$ is a Modular Form of weight k if:*

1. *f is holomorphic on $\mathbb{H}$*
2. *$f(\tau)$ bounded as $im(\tau) \to \infty$ (bounded along the y coordinate of the complex plane)*
3. *$f(\tau + 1) = f(\tau)$ and $f(-1/\tau) = (\tau)^k f(\tau)$ $\forall \tau \in \mathbb{H}$*

Corollary 1.2.11. *Since f is $\mathbb{Z}$ periodic, and holomorphic, we can fourier expand f, which gives*

$$f(x + iy) = \sum_{n \in \mathbb{Z}} a_n e^{2\pi i n(x+iy)} = \sum_{n \in \mathbb{Z}} a_n e^{2\pi i n x} e^{-2\pi n y}$$

*If a_n for negative n were to be non zero, we would have that f is unbounded as $y \to \infty$ (since $e^{2\pi n y}$ would go unbounded). Therefore, $\forall n < 0, a_n = 0$. a_0 can be 0, in which case $\lim_{y \to \infty} f(x + iy) = a_0 = 0$. If $a_0 = 0$, then the modular form is called a **Cusp Form**.*

Weakly Holomorphic Modular Forms of weight k (for $SL_2(\mathbb{Z})$)

Definition 1.2.12. **(Weakly Holomorphic Modular Forms of weight k)** We say a function $f : \mathbb{H} \to \mathbb{C}$ is a weakly holomorphic modular form of weight k if:

1. $f(a\tau + b/c\tau + d) = (c\tau + d)^k f(\tau)$ for every $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2, \mathbb{Z})$
2. $f(\tau) \in \mathcal{O}(q^{-N})$ instead of $\mathcal{O}(q^0)$ (for $f = \sum_{n \in \mathbb{Z}} a_n q^n$) which yields $f = \sum_{n=-N}^{\infty} a_n q^n$ (Note that f loses the boundedness property as $im(x)$ goes to infinity.)

Definition 1.2.13. **(M_k)** Space of all Modular Forms of weight k (Forms a vector space over $\mathbb{C}$).

Definition 1.2.14. **(S_k)** Space of all Cusp forms ($a_0 = 0$, i.e. $f(x + iy) \to 0$ as $y \to \infty$) (Forms a vector space over $\mathbb{C}$).

Definition 1.2.15. **($M_k^!$)** Space of all weakly holomorphic forms (Forms a vector space over $\mathbb{C}$).

Fact 1.2.16.

$$S_k \subset M_k \subset M_k^!$$

Remark 1.2.17. Note, it is more prudent to specify the group for which we are discussing these Modular Forms, for eg. $S_k(SL_2(\mathbb{Z}))$ or $M_k(SL_2(\mathbb{Z}))$ (for reasons that will be clear later).

Some Growth Properties of Modular forms ($SL_2(\mathbb{Z})$)

1. For $f \in S_k(SL_2(\mathbb{Z}))$, $a_n \in \mathscr{O}(n^{k/2})$
2. For $f \in M_k(SL_2(\mathbb{Z})), a_n \in \mathscr{O}(n^{k-1})$
3. For $f \in M_k^!(L_2(\mathbb{Z})), a_n \in \mathscr{O}(e^{C\sqrt{n}})$

Definition 1.2.18. **(Eisenstein Series E_k)**

$$E_k(z) := \sum_{(m,n)\in\mathbb{Z}^2,(m,n)\neq 0} \frac{1}{(mz+n)^k}$$

Remark 1.2.19. $E_k(z)$ is a modular form of weight k (for $SL_2(\mathbb{Z})$) for all $k \geq 4$. They are very important examples of modular forms for they, in some sense, form the "building blocks" to construct other modular forms.

Definition 1.2.20. **(Discriminant function Δ)**

$$\Delta(z) := q\Pi_1^\infty(1-q^n)^{24}$$

(Where $q = e^{2\pi i z}$). This function is a *cusp form* of weight 12.

We present, without proof, one of the cornerstone theorems in Modular Forms theory:

Theorem 1.2.21. *1. $M_k(SL_2(\mathbb{Z}))$ is generated by the monomials $E_4^\alpha E_6^\beta$ such that $4\alpha+6\beta = k$. In other words, as a vector space we see that*

$$M_k(SL_2(\mathbb{Z})) = \langle\{E_4^\alpha E_6^\beta : 4\alpha+6\beta = k \text{ (in integers)}\}\rangle$$

A trivial corollary of this is that, $M_k(SL_2(\mathbb{Z}))$ is a finite dimensioned vector space.

2.

$$M_*(SL_2(\mathbb{Z})) = \oplus_k M_k(SL_2(\mathbb{Z}))$$

forms a Graded Algebra- *A direct sum of submodules that forms an algebra. Moreover,*

$$M_*(SL_2(\mathbb{Z})) = \mathbb{C}[E_4, E_6]$$

This says that, in principle, the space of all modular forms for $SL_2(\mathbb{Z})$ acts as the set of all 2 variable polynomials over $\mathbb{C}$ (E_4 and E_6 are algebraically independent).

3.

$$M_k(SL_2(\mathbb{Z})) = \mathbb{C}\cdot E_k + S_k$$

This result says that every modular form is some scalar times the Eisenstein, with the "error term" being a cusp form (or the other way around, where every modular form is a cusp form, with the error term being a factor of the Eisenstein)

4.

$$S_k = \Delta \cdot M_{k-12}$$

Remark 1.2.22. With the results 3 and 4 of 1.2.21, we see that $M_k = \mathbb{C}E_k + \Delta M_{k-12} = \mathbb{C}E_k + \Delta(\mathbb{C}E_{k-12} + \Delta M_{k-24}) = \mathbb{C}E_k + \Delta(\mathbb{C}E_{k-12} + \Delta(\mathbb{C}E_{k-24} + \Delta M_{k-36})) = \cdots$ from which we get that any modular form f of weight k can be written as

$$f = \sum_{n=0}^{[k/12]} \alpha_n E_{k-12n} \Delta^n$$

1.2.2 Fundamental Domain and the action of $SL_2(\mathbb{Z})$ on $\mathbb{H}$

The defining action of a $\gamma = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in SL_2(Z)$ on a vector $z \in \mathbb{H}$ is given by:

$$z \mapsto \gamma z = \frac{az+b}{cz+d}$$

Note the following fact:

$$\operatorname{im}(\frac{az+b}{cz+d}) = \frac{(ad-bc)\operatorname{im}(z)}{|cz+d|^2}$$

(This, in some sense, necissitates the need for us to look at $GL_2^+(\mathbb{R})$. The necessity to look at $SL_2(\mathbb{Z})$ comes from analysing Lattices of the form $\omega_1\mathbb{Z} \oplus \omega_2\mathbb{Z}$ (for $\omega_1, \omega_2 \in \mathbb{R}$). A change of basis of the lattice corresponds to a $SL_2(\mathbb{Z})$ multiplication).
Since $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \in SL_2(\mathbb{Z})$, the respective actions are $z \mapsto z+1$ (translations) and $z \mapsto -1/z$ (inversions).

Definition 1.2.23. **(Fundamental Domain $\mathcal{F}$ for $\Gamma \leq SL_2(\mathbb{Z})$)** A fundamental domain is a subset of $\mathbb{H}$ under the action of Γ, i.e.

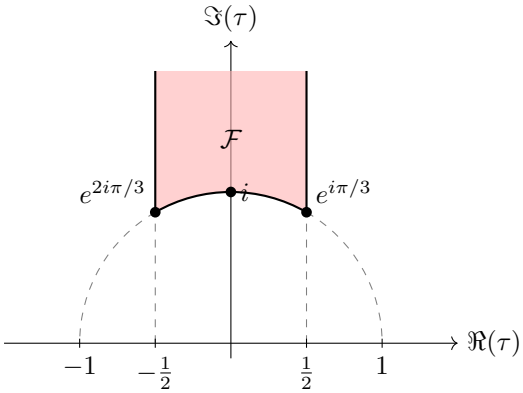
$$\mathbb{H} = \cup_{\gamma \in \Gamma} \gamma \mathcal{F}$$

such that:

1. $(\gamma_1 \mathcal{F})^\circ \cap (\gamma_2 \mathcal{F})^\circ = \varnothing$.
2. $(\bar{\mathcal{F}})^\circ = \mathcal{F}$ (The set is "full" in some sense).
3. $\forall z \in \mathbb{H}$, $\exists d \in \mathcal{F}$ and $\gamma \in \Gamma$ such that $\gamma d = z$.

Example 1.2.24. A fundamental Domain for $SL_2(\mathbb{Z})$ is

$$\mathcal{F}_{SL_2(\mathbb{Z})} := \{x \in \mathbb{H} : |\operatorname{Re}(x)| \leq \frac{1}{2} \text{ and } |x| > 1\}$$



Shimura's Slash Notation

Definition 1.2.25. $\big(f(z)|_k[\gamma]\big)$ Let $\gamma \in GL_2^+(\mathbb{Q})$ Define

$$f(z)|_k[\gamma] := (\det(\gamma))^{k/2} j(\gamma, z) f(\gamma z)$$

Where $\gamma z := \frac{az+b}{cz+d}$ and $j(\gamma, z) := (cz+d)^{-k}$ (Called the Automorphy factor). Altogether the above shorthand reads:

$$f(z)|_k[\gamma] := (\det(\gamma))^{k/2} (cz+d)^{-k} f\left(\frac{az+b}{cz+d}\right)$$

Fact 1.2.26. $f(z)|_k[\gamma_1\gamma_2] = (f(z)|_k[\gamma_1])|_k[\gamma_2]$

What are Cusps?

We begin by considering the map

$$z = x + iy \mapsto q = e^{2\pi i z} = e^{2\pi i x} e^{-2\pi y}.$$

Since $y > 0$, we have $0 < |q| < 1$, so this map sends the upper half-plane $\mathbb{H}$ to the punctured unit disk.

As $y \to \infty$, we see that $|q| = e^{-2\pi y} \to 0$. Thus the point $z = i\infty$ corresponds to $q = 0$. On the other hand, as $y \to 0^+$, we have $|q| \to 1$, so points approach the boundary of the unit disk.

This shows that the upper half-plane $\mathbb{H}$ corresponds to a punctured disk, with the missing point $q = 0$ corresponding to $z = i\infty$.

Now consider first the action of $SL_2(\mathbb{Z})$ on $\mathbb{H}$. The quotient space $SL_2(\mathbb{Z})\backslash\mathbb{H}$ can be visualized using a fundamental domain $\mathcal{F} \subset \mathbb{H}$, for example:

$$\mathcal{F} = \left\{ z \in \mathbb{H} : |\operatorname{Re}(z)| \leq \tfrac{1}{2},\ |z| \geq 1 \right\}.$$

This space, however, is non compact. In order to compactify this space, (and viably include the relevant points, i.e. those that correspond to $q = 0$ and $|q| = 1$)

Appendices

Appendix A

Modular Curves and Cusps

A.1 The Problem of Non-Compactness

Modular forms for $SL_2(\mathbb{Z})$ live naturally on the upper half-plane $\mathbb{H}$, where the fundamental domain $\mathcal{F}$ captures the geometry of the quotient $SL_2(\mathbb{Z})\backslash\mathbb{H}$[cite: 1]. However, $\mathcal{F}$ is not compact; it possesses a non-compact "end" as $\mathrm{Im}(z) \to \infty$[cite: 2]. This non-compactness prevents the direct application of standard results from the theory of compact Riemann surfaces, such as the Riemann-Roch theorem[cite: 3].

Furthermore, the q-expansion $f(z) = \sum_{n=0}^{\infty} a_n q^n$ of a modular form has a "missing point" at $q = 0$, which corresponds to $z = i\infty$[cite: 4]. The boundedness condition in the definition of a modular form is essentially a requirement on the behavior at this missing point[cite: 5]. When generalizing to congruence subgroups $\Gamma \leq SL_2(\mathbb{Z})$, multiple non-compact ends appear, each contributing a missing point known as a *cusp*[cite: 6, 7].

A.2 The Boundary of $\mathbb{H}$ and the Rational Projective Line

Topologically, the boundary of $\mathbb{H}$ in $\mathbb{C}$ is $\mathbb{R} \cup \{i\infty\} \cong S^1$[cite: 8]. A naive compactification adjoining all of $\mathbb{R} \cup \{i\infty\}$ to $\mathbb{H}$ fails because the action of $SL_2(\mathbb{Z})$ on irrational points in $\mathbb{R}$ is dense and non-discrete, leading to a non-Hausdorff quotient space[cite: 9, 10, 19].

Conversely, $SL_2(\mathbb{Z})$ preserves the set of rational points $\mathbb{Q} \cup \{i\infty\}$[cite: 12].

Definition A.2.1. **(Projective Line over $\mathbb{Q}$)**

$$\mathbb{P}^1(\mathbb{Q}) := \mathbb{Q} \cup \{i\infty\}$$

We represent a rational p/q in lowest terms as $[p : q] \in \mathbb{P}^1(\mathbb{Q})$, with $i\infty$ as $[1 : 0]$[cite: 25]. The action of $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$ is given by $\gamma \cdot [p : q] = [ap + bq : cp + dq]$[cite: 26].

Lemma A.2.2. ***Transitivity of the Action*** *$SL_2(\mathbb{Z})$ acts transitively on $\mathbb{P}^1(\mathbb{Q})$[cite: 27].*

Proof. For any $[p : q] \in \mathbb{P}^1(\mathbb{Q})$ with $\gcd(p, q) = 1$, we seek $\gamma \in SL_2(\mathbb{Z})$ such that $\gamma \cdot (p/q) = i\infty$, implying the denominator $cp + dq = 0$[cite: 28, 29]. Setting $c = q$ and $d = -p$, we

have $\gcd(c, d) = 1$[cite: 30]. By Bezout's theorem, there exist a, b such that $ad - bc = 1$, thus $\gamma = \begin{pmatrix} a & b \\ q & -p \end{pmatrix} \in SL_2(\mathbb{Z})$ sends p/q to $i\infty$[cite: 31].

□

Corollary A.2.3. *$SL_2(\mathbb{Z})$ has exactly* ***one*** *cusp, as $\mathbb{P}^1(\mathbb{Q})$ is a single orbit that collapses to one point in the quotient[cite: 32].*

Definition A.2.4. **(Cusps of Γ)** Let $\Gamma \leq SL_2(\mathbb{Z})$ be a subgroup. The *cusps of* Γ are the orbits of the action of Γ on $\mathbb{P}^1(\mathbb{Q})$:

$$\mathrm{Cusps}(\Gamma) := \Gamma \backslash \mathbb{P}^1(\mathbb{Q})$$

For a finite-index subgroup, there are finitely many such cusps[cite: 34, 35].

A.3 The Horoball Topology and Modular Curves

To compactify the quotient, we define the *horoball topology* on $\mathbb{H}^* := \mathbb{H} \cup \mathbb{P}^1(\mathbb{Q})$[cite: 36, 41].

Definition A.3.1. **(Horoball Neighborhoods)**

1. For $z_0 \in \mathbb{H}$, neighborhoods are standard open sets in $\mathbb{C}$[cite: 38].
2. For the cusp $i\infty$, a neighborhood is $U_M(i\infty) := \{z \in \mathbb{H} : \mathrm{Im}(z) > M\} \cup \{i\infty\}$ for $M > 0$[cite: 39].
3. For a finite cusp $s \in \mathbb{Q}$, let $\sigma \in SL_2(\mathbb{Z})$ such that $\sigma(i\infty) = s$. A neighborhood is $U_M(s) := \sigma\left(U_M(i\infty)\right) \cup \{s\}$[cite: 40].

Definition A.3.2. **(Modular Curve $X(\Gamma)$)** For a finite-index subgroup $\Gamma \leq SL_2(\mathbb{Z})$, the modular curve is the quotient:

$$X(\Gamma) := \Gamma \backslash \mathbb{H}^* = \Gamma \backslash \left(\mathbb{H} \cup \mathbb{P}^1(\mathbb{Q})\right)$$

equipped with the quotient topology from the horoball topology[cite: 45, 46].

Theorem A.3.3. ***Compactness of the Modular Curve*** *$X(\Gamma)$ is a compact Riemann surface[cite: 47].*

Proof. The non-compact ends of $\Gamma \backslash \mathbb{H}$ correspond to the finitely many cusps of Γ[cite: 48]. Each cusp horoball maps to a punctured disk around $q = 0$ in local coordinates[cite: 49]. Adjoining these finitely many points fills the punctures, resulting in a compact space[cite: 50]. At a cusp s with width h, the map $z \mapsto q_h = e^{2\pi i z/h}$ provides a well-defined complex chart[cite: 52, 53].

□

A.4 Cusps and q-Expansions

Definition A.4.1. **(Width of a Cusp)** Let s be a cusp of Γ and $\sigma \in SL_2(\mathbb{Z})$ satisfy $\sigma(i\infty) = s$. The *width* h is the smallest positive integer such that $\sigma^{-1}\begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix}\sigma \in \Gamma$[cite: 54, 55].

Corollary A.4.2. *For $f \in M_k(\Gamma)$, each cusp s of width h yields a convergent expansion $f(z) = \sum_{n=0}^{\infty} a_n(s)\, q_h^n$. For a cusp form $f \in S_k(\Gamma)$, we require $a_0(s) = 0$ at* ***every*** *cusp s[cite: 56].*

Remark A.4.3. In the context of the Dabholkar–Murthy–Zagier paper, this philosophy extends to Jacobi and Siegel modular forms[cite: 64]. The cusps of the resulting compactified spaces index the different q-expansions of the modular objects, and being a cusp form necessitates vanishing at all such boundary points[cite: 66, 67].